

Astro goal

TDP
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Overview

Our code requires `numpy` library to run

Our 1st Robot is the Goal-Keeper

The other two are attackers and sometimes they do defense

2nd and 3rd Robot send ball and robot global data to 1st Robot

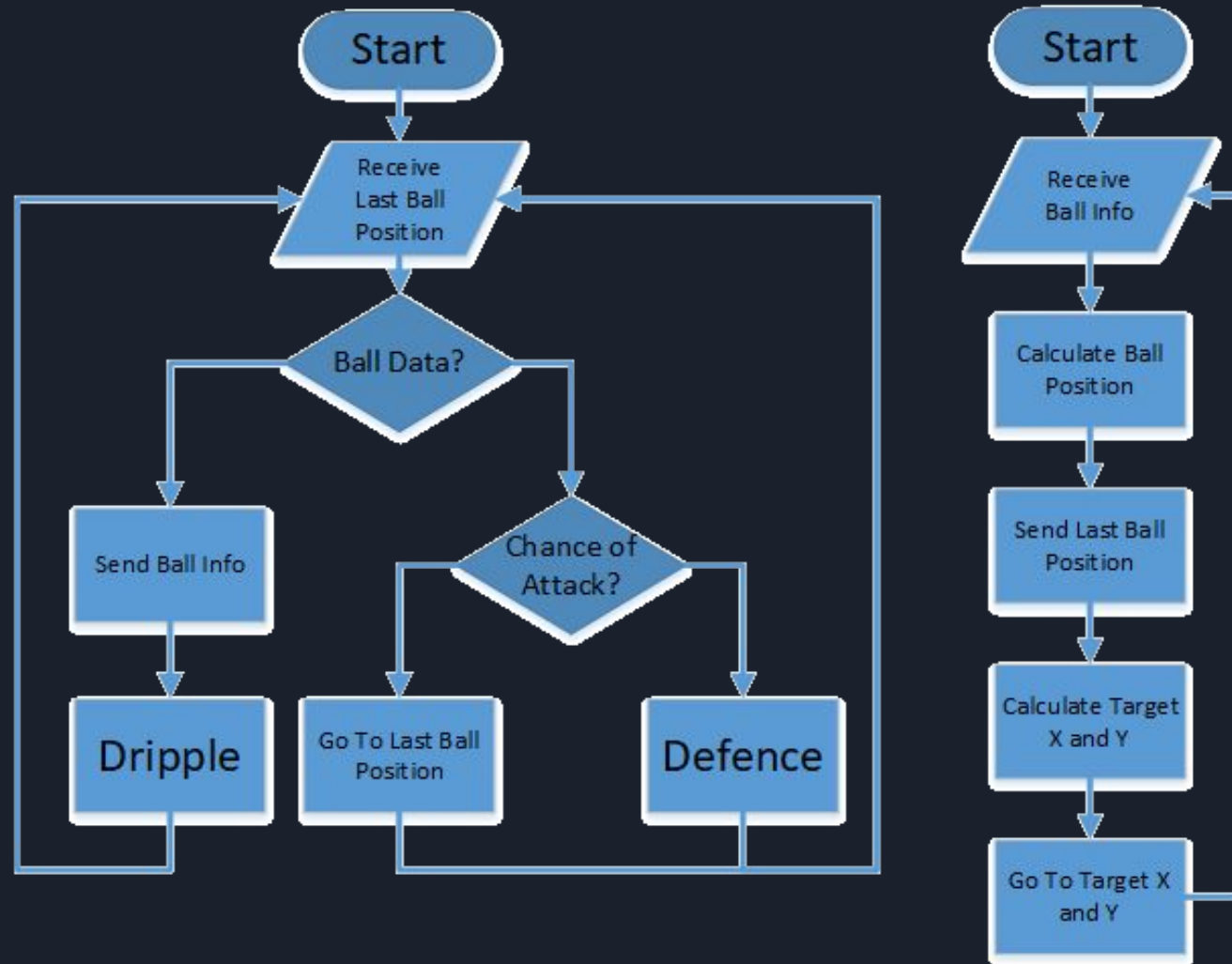
1st Robot calculates ball position and sends it to the team

If there are 2 ball data a system of equations is solved using numpy

If there is only 1/no ball data its position is estimated/predicted

We utilized the rotation matrix a few times in our code

Flow Charts



To Do's

A more complex Algorithm to work better as a team

Having a dynamic Goal-Keeper instead of a fixed one

Detect Other team robots and track their position

A more advanced motion-planning

Integrating an effective AI model

Thank you for reading



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